Take-Home Exam, Advanced Microeconomics — v2

Problem 1: Overworked Americans – Q1, Q2, Q4, Q5, Q7

Alessandro Caggia (), Stefano De Bianchi (), Eduard (), Francesco Mastidoro (3323853)

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1 Q1 – The hours gap and its noisy relation to tax wedges

Question 1. *Document the fact described above about the difference between hours worked in the US vs Europe in recent years, if needed even across different European countries.*

1.1 Definitions

For the working-age (15–64) population: hours per employed worker HE (the intensive object the labour-supply model speaks to), employment rate $e \equiv E/N$, hours per adult $H/N \equiv HE \cdot e$. The Bick–Brüggemann–Fuchs-Schündeln (2019) decomposition $\Delta \log(H/N) = \Delta \log HE + \Delta \log e$ assigns roughly half of the EU–US gap to each margin; we focus on HE.[1]

1.2 The fact

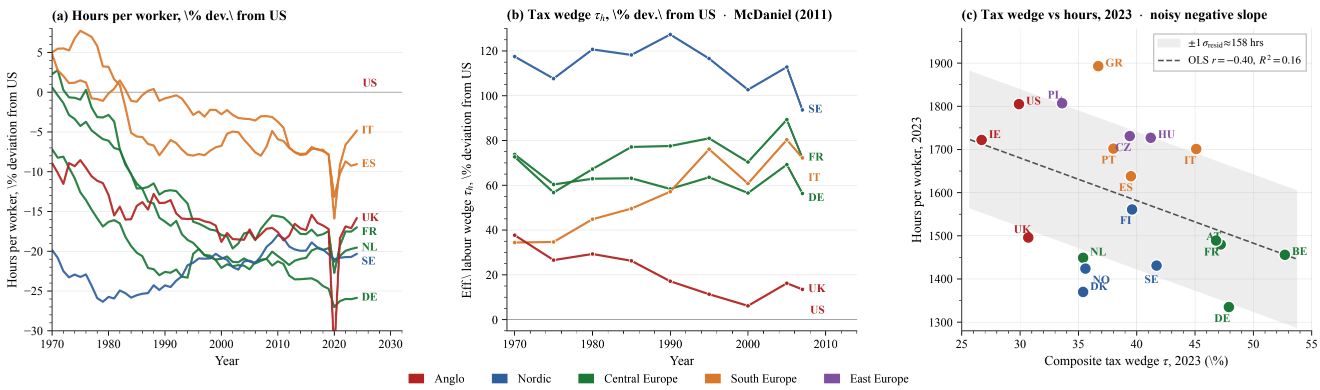

Figure 1: **Three-panel evidence.** (a) Hours per worker, % deviation from US, 1970–2024 (OECD). (b) Effective labour wedge τ_h, % deviation from US, 1970–2007. (c) OECD labour-tax wedge vs hours per worker, 2023 cross-section ($n = 19$); OLS line and $\pm 1\sigma_{\text{resid}}$ band. *Source:* `research/build_figure_panel.py`.

Three readings. **(i)** Hours fell in every focus country relative to the US since 1970 (DE/NL ≈ -25 to -30%; IT/ES ≈ -10 to -15%). **(ii)** The cross-country labour-wedge gap to the US has been roughly stable since 1970, so a tax-driven story has to rationalise a roughly time-invariant cross-country hours gap — broadly consistent with panel (a). **(iii)** The 2023 cross-section is signed correctly but very noisy: OLS slope -9.9 hrs/pp, $R^2 = 0.16$, residual SD ≈ 158 hours. The entire DE–US gap is ≈ 470 hours, so the residual SD is one third of what we want to explain. Italy ($\tau = 45.1\%$) and Greece (36.7%) work *more* than France (46.8%); the four highest-τ countries (BE, DE, AT, FR) span ≈ 200 hours within a 6 pp wedge band.

1.3 The inconsistency the rest of the problem is asked to explain

Taxation alone fits poorly: $R^2 = 0.16$ and a residual SD of one third of the gap to explain. Q2 asks whether a structural labour-supply model can recover the tax-elasticity number that this fit would require, and whether that number is consistent with microeconometric evidence. Q4 and Q5 propose two institutional channels that could absorb the residual. We treat home production / time-use measurement as outside the scope of this submission.

[1] The extensive margin is shut down throughout this submission — a limitation we acknowledge rather than silently impose. Cross-country e differences are substantial (US 71% vs IT 59% in 2019). The prompt asks about hours, so HE is the relevant object; the participation channel would, if anything, push our calibrated α in Q2 *upward* and so deepen the residual we identify.

2 Q2 – Prescott model and the elasticity that would be needed

Question 2. *One explanation that has been put forward — for example, by Ed Prescott in 2004 ('Why do Americans work so much more than Europeans?') — is higher taxes. Summarize the main differences in the income tax between US and Europe (or a few European countries). Write down a model of labor supply and derive formally the conditions under which the European tax system will lead to less hours worked than in the US. [Hint: start with the simplest case of differences only in the average tax rates.]*

2.1 Modelling choice

We use log–log preferences with a balanced lump-sum rebate (Prescott 2004). *What this buys.* A one-parameter (α) closed-form $h^*(\tau)$ and a clean intensive-elasticity comparative static — the simplest setup in which the question "which ε would the model need to fit the data?" has a sharp answer. *What it costs.* The model is static (no Frisch margin), balanced-budget (income effects rebated lump-sum), and abstracts from capital and worker heterogeneity. Heterogeneity returns in Q4; the extensive margin and home production are shut down (footnote, Q1).

2.2 The simplest one-margin model

Log–log preferences $u(c,h) = \log c + \alpha \log(1-h)$, wage w, proportional labour-tax wedge τ, lump-sum rebate T. Worker budget is $c = (1-\tau)wh + T$ with T rebated such that $c = wh$ in equilibrium. The static FOC is the canonical $(1-\tau)w\, u_c = -u_l$:

$$\frac{(1-\tau)w}{c} = \frac{\alpha}{1-h}, \qquad c = wh \implies \frac{1-\tau}{h} = \frac{\alpha}{1-h} \implies h^*(\tau) = \frac{1-\tau}{\alpha+1-\tau}, \qquad \varepsilon^M_{h,1-\tau} = \frac{\alpha}{\alpha+1-\tau} \in (0,1).$$

$\partial h^*/\partial\tau < 0$, so the model delivers a sharp formal condition for "Europe works less than the US" under common preferences:

$$h^*_E < h^*_U \iff \tau_E > \tau_U \qquad \text{(at common } \alpha).$$

Remark. With country-specific α (different leisure-disutility weights), the condition is no longer monotone in τ alone: $h^*_E < h^*_U$ becomes joint in (τ,α), and Europe could work less even at common τ if $\alpha^{EU} > \alpha^{US}$ (i.e., Europeans put higher weight on leisure in the utility function). This is the channel Q5 (unions) and Q7 (λ) reactivate via institutions and welfare weights rather than via preferences directly. The elasticity ε^M is *Marshallian* (uncompensated, holds the rebate fixed at the margin), not Frisch — the static model has no intertemporal margin.

2.3 Anchor and the cross-section fit

Anchoring $h^{US} = 0.33$ at $\tau^{US} = 0.35$, invert $h^*(\tau) = (1-\tau)/(\alpha+1-\tau)$ for α:

$$\alpha = \frac{(1-\tau)(1-h^*)}{h^*} = \frac{0.65 \cdot 0.67}{0.33} = 1.32.$$

The implied Marshallian intensive elasticity at the US point is $\varepsilon^M = \alpha/(\alpha+1-\tau) = 1.32/(1.32+0.65) = 0.67$, rising along the supply curve to 0.75 at $\tau = 0.56$. Predicted EU hours at $\tau^{EU} = 0.56$: $h^{EU}_{\text{pred}} = (1-0.56)/(1.32+1-0.56) = 0.44/1.76 = 0.250$ vs. data 0.247. *The calibrated model essentially fits the cross-section, but at an α inconsistent with quasi-experimental evidence.* The substantive content of the controversy is precisely the gap between the implied $\varepsilon^M \in [0.67, 0.75]$ and the micro consensus ≈ 0.30.

2.4 Decomposition at the micro elasticity

At $\alpha = 1.32$, α absorbs everything (preferences, institutions, culture). Restricting α to be micro-consistent isolates the share of the gap that the Prescott mechanism can carry on its own. Solve $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.30$ at $\tau^{US} = 0.35$:

$$\alpha^{\text{micro}} = \frac{0.30 \cdot (1-\tau)}{1-0.30} = \frac{0.30 \cdot 0.65}{0.70} = 0.279.$$

Predicted hours at this α: $h^*_{US} = 0.65/(0.279 + 0.65) = 0.700$, $h^*_{EU} = 0.44/(0.279 + 0.44) = 0.612$, log-drop $\log(0.612/0.700) = -0.134$. Data log-drop is -0.290. Hence:

$$\text{Prescott share of EU–US hours gap} = \frac{0.134}{0.290} \approx 46\%, \qquad \text{residual} \approx 15.6\,\text{pp} \approx 54\%.$$

Taxes alone explain about 46% **of the observed gap; the residual** 54% **lies outside the Prescott mechanism, and its source is not yet identified.** Q4 (hours regulation) and Q5 (unions) propose two candidate institutional channels and ask whether each can absorb part of this residual.

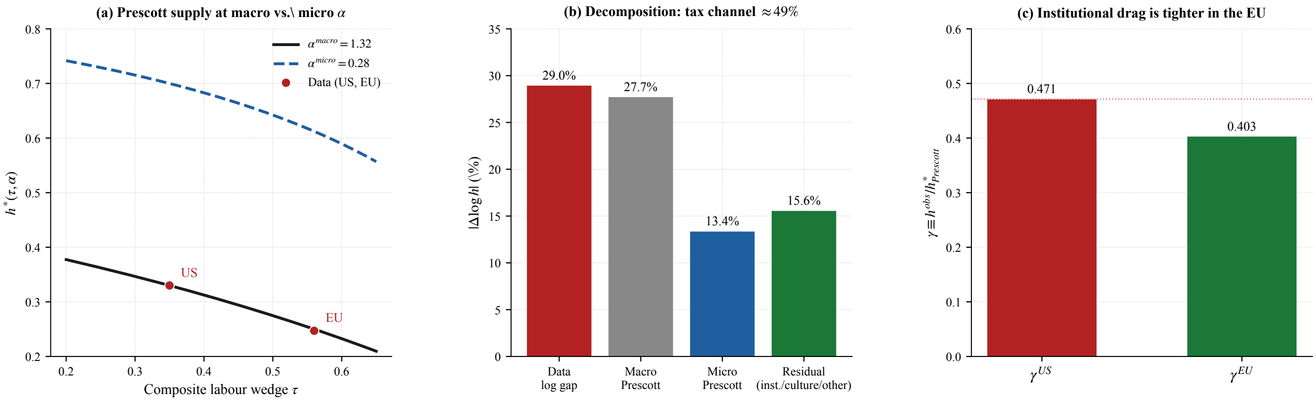

Figure 2: **Q2 macro–micro decomposition.** **(a)** Prescott supply curves at $\alpha^{\text{macro}} = 1.32$ and $\alpha^{\text{micro}} = 0.279$; US/EU data marked. **(b)** Decomposition of the 29% data gap: Prescott-at-micro explains 13.4% (46%); unexplained residual 15.6% (54%). **(c)** Unexplained-shortfall ratio $\gamma \equiv h^{\text{obs}}/h^*_{\text{Prescott}}(\alpha^{\text{micro}})$: cross-country ratio $\gamma^{EU}/\gamma^{US} = 0.858$, i.e. EU loses about 14% extra hours relative to US, beyond Prescott-at-micro — the precise quantity Q4 / Q5 must produce. *Source:* `research/sim_models_v2.py`.

3 Common framework for Q4 and Q5

3.1 Primitives

We adopt a quadratic-disutility worker, linear consumption, and a linear-quadratic firm-profit function — the simplest closed-form specification that admits both worker- and firm-preferred hours and sign-determinate welfare conditions:

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2, \;\; h^* = \frac{(1-\tau)w}{\alpha}; \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh, \;\; h^{\text{firm}} = \frac{A-w}{\beta}.$$

Joint surplus is $W(h) \equiv V_w(h) + V_f(h) = Ah - \tau wh - \frac{\alpha+\beta}{2}h^2$. The wage transfer cancels between worker and firm; only the tax wedge τwh survives. W is concave with maximum

$$h^{\text{soc}}(\tau, \alpha, A, \beta, w) = \frac{A - \tau w}{\alpha + \beta}.$$

The setup is shared by Q4 (regulation) and Q5 (unions). Wages are partial-equilibrium; heterogeneity returns at the end of Q4.

3.2 Baseline calibration and the firm-power ordering

$A = 2$, $w = 1$, $\alpha = \beta = 2$, $\tau = 0.4$ gives $h^* = 0.30$, $h^{\text{soc}} = 0.40$, $h^{\text{firm}} = 0.50$. **Justification of the parameter choices.** (i) $\tau = 0.4$ is the population-weighted midpoint between US (0.35) and EU (0.56) labour-tax wedges (OECD *Taxing Wages 2023*, Q1 panel (c)). (ii) The hours scale anchors $h^* = 0.30$ to BLS-OECD hours-per-worker over awake-time at the US point ($\sim$2,000 work hours / $\sim$6,500 awake hours ≈ 0.31, in line with Prescott 2004). (iii) $\alpha = \beta$ is a symmetry assumption that makes the welfare-improving band of (1) collapse to the textbook expression (h^*, h^{firm}), but **only the ordering** $h^* < h^{\text{soc}} < h^{\text{firm}}$ **is load-bearing** for what follows; the qualitative results are invariant in α, β separately. (iv) $A = 2$, $w = 1$ are normalisations: the level of A is identified only relative to w, and our results depend on the ratio A/w, not the absolute scale.

The ordering says that, when the firm has all the bargaining power, it overshoots the social optimum. We refer to this as the *firm-power baseline* (the firm sets hours on its labour-demand curve at the wage that makes the worker indifferent between working and her outside option). With identical τ and identical preferences, the model predicts no US–EU hours gap under firm power *or* under union bargaining: the gap must come from a country-specific institutional asymmetry, which Q4 (cap) and Q5 (union strength) supply.

4 Q4 – Hours regulation: the optimality condition for a cap

Question 4. *Another explanation that has been put forward is labor market regulation limiting the number of hours that Europeans can work. The impact of such regulation is obvious, but the welfare effects are more subtle. Can you derive the welfare effects of such labor legislation on workers and firms? How would your answer change if workers are heterogeneous?*

4.1 Modelling choice

A cap $\bar{h}$ is a quantity instrument that takes the firm's labour-demand response as given. *What the trade-off captures.* Workers value leisure (low h^*); firms value output (high h^{firm}); a binding cap forces hours from h^{firm} toward $\bar{h}$. Whether the redistribution is welfare-improving depends on whether $\bar{h}$ lands inside or outside the welfare-improving band defined by joint-surplus concavity: too tight a cap overshoots and destroys joint surplus. *What this buys.* A closed-form welfare condition that depends only on the gap $h^{\text{firm}} - h^{\text{soc}}$, transparent in the heterogeneity extension. *What it costs.* (a) Extensive margin shut down (Q1 footnote): no firm response on number of bodies hired. (b) Wages held at the firm-power IR-binding level (no adjustment of w to absorb the cap). (c) Heterogeneity on α only (workers differ in disutility curvature; we abstract from heterogeneous wages and outside options). The closest sufficient-statistics analogue is Lee–Saez (2012) on the minimum wage, but their welfare logic runs through extensive-margin rationing whereas a binding hours cap rations *intensive*-margin only.

4.2 Setup: firm-power baseline

At $w^* = 1$ pinning the worker's IR at V_w^0, the firm picks $h^{\text{firm}} = (A - w^*)/\beta = 0.50$. Hours sit *above* $h^{\text{soc}} = 0.40$, so the cap has something to correct. With cap $\bar{h}$ the constrained equilibrium is $h^{\text{eq}}(\bar{h}) = \min\{\bar{h},\, h^{\text{firm}}\}$.

4.3 Worker, firm, and joint surplus under a cap

The three surpluses, evaluated at $h^{\text{eq}}(\bar{h})$ relative to the no-cap baseline, are quadratic in $\bar{h}$ around the respective bliss points:

Surplus	Closed form (for $\bar{h} < h^{\text{firm}}$)	Sign
$\Delta V_w(\bar{h}) = V_w(\bar{h}) - V_w(h^{\text{firm}})$	$(\bar{h} - h^{\text{firm}})\big[(1-\tau)w - \frac{\alpha}{2}(\bar{h} + h^{\text{firm}})\big]$	> 0 iff $\bar{h} \in (2h^* - h^{\text{firm}},\, h^{\text{firm}})$
$\Delta V_f(\bar{h}) = V_f(\bar{h}) - V_f(h^{\text{firm}})$	$-\frac{\beta}{2}(h^{\text{firm}} - \bar{h})^2$	< 0 for any $\bar{h} < h^{\text{firm}}$
$\Delta W(\bar{h}) = \Delta V_w + \Delta V_f$	$-\frac{\alpha+\beta}{2}\big[(h^{\text{soc}} - \bar{h})^2 - (h^{\text{soc}} - h^{\text{firm}})^2\big]$	> 0 iff $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}})$

4.4 Optimality condition (the Q4 result)

Claim. Joint surplus $W(h)$ is strictly concave and quadratic with peak at h^{soc}; hence $W(\bar{h})$ is symmetric in $\bar{h}$ around h^{soc}. The function $\Delta W(\bar{h})$ thus equals zero at $\bar{h} = h^{\text{firm}}$ (no cap) and at the mirror point $2h^{\text{soc}} - h^{\text{firm}}$, and is positive between them. Hence:

$$\Delta W(\bar{h}) \begin{cases} = 0 & \text{if } \bar{h} \geq h^{\text{firm}} \\ > 0 & \text{iff } \bar{h} \in \left(2h^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}}\right) \\ < 0 & \text{if } \bar{h} < 2h^{\text{soc}} - h^{\text{firm}} \end{cases} \tag{1}$$

provided $h^{\text{firm}} > h^{\text{soc}}$ (which holds at our baseline, $0.50 > 0.40$). The peak gain at $\bar{h} = h^{\text{soc}}$ is $\frac{1}{2}(\alpha + \beta)(h^{\text{firm}} - h^{\text{soc}})^2 = 0.020$ at calibration. At $\alpha = \beta$ the band collapses to $\bar{h} \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$.

Reading. A cap that pulls h *toward* the joint-surplus optimum is unambiguously welfare-improving. A cap that pulls h *below* h^{soc} overshoots, and once $\bar{h} < 2h^{\text{soc}} - h^{\text{firm}} = h^* = 0.30$ even the worker is hurt — joint surplus turns negative.

4.5 Heterogeneous workers: who is rationed

With worker types indexed by the leisure-disutility curvature α_i (but identical w, τ), the cap forces every type to $\bar{h}$. The individual payoff change $\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})\big[(1-\tau)w - \frac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})\big]$ is positive iff $\alpha_i > \alpha^*(\bar{h})$ where

$$\alpha^*(\bar{h}) = \frac{2(1-\tau)w}{\bar{h} + h^{\text{firm}}}. \tag{2}$$

At $\bar{h} = h^{\text{soc}} = 0.4$: $\alpha^* = 1.333$. **High-α (leisure-loving) workers gain; low-α (income-loving) workers are rationed on the intensive margin**: their preferred hours $h_i^* = (1-\tau)w/\alpha_i$ exceed $\bar{h}$ and the cap forces them below their personal optimum. The cap is therefore a redistributive instrument across worker types: it transfers welfare from low-α workers to high-α workers and away from the firm.

Aggregate condition. Integrating over the type distribution F with mean $\bar{\alpha}$, the cap is welfare-improving on average iff

$$\int \Delta V_w^i(\bar{h})\, dF(\alpha) + \Delta V_f(\bar{h}) > 0.$$

At baseline ($\bar{\alpha} = 2$, $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$) the cap is welfare-improving on average; a workforce with median $\alpha = 1$ would lose.

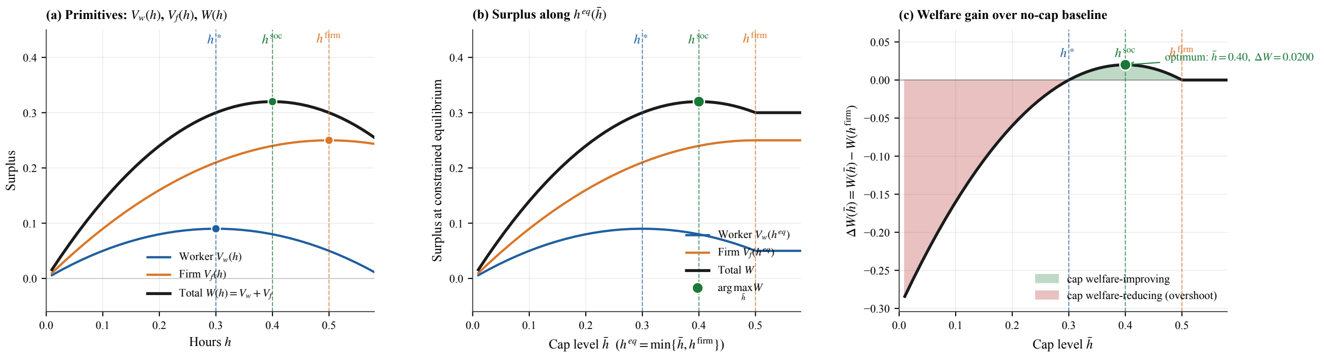

Figure 3: **Hours-cap simulation. (a)** Primitives V_w, V_f, W over h with bliss points marked; **(b)** surpluses along $h^{\text{eq}}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$; **(c)** $\Delta W(\bar{h})$ vs no-cap baseline — green band is the welfare-improving region of (1), red is overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. *Source:* `research/sim_mechanisms.py`.

Empirical anchor. The 35-hour Aubry reform (France 1998–2000) documented by Estevão–Sá (2008) found $\approx 10-13\%$ hours cut and near-zero employment effect, consistent with a binding cap at $\bar{h}/h^{\text{firm}} \approx 0.875$ inside the welfare-improving band of (1). The empirical near-zero employment effect is consistent with our model only because we have shut down the extensive margin (Q1 footnote): if a binding cap reduces hours per worker, firms may substitute workers for hours, an effect the representative-agent setup rules out by construction. Sectoral composition (a large public sector with multi-year collective contracts) is also outside the model.

5 Q5 – A single union as welfare-restorer in a two-sector economy

Question 5. *Yet a third explanation is the presence of unions in the economy and in particular how unions respond to sectoral shifts in the economy. Show the impact of sectoral shifts (i.e. one part of the economy gets more productive and the other less so) on hours worked in a competitive economy. Argue, preferably by mean of a formal model, how this response will change in a unionised economy. What data would you collect to see whether your theoretical hypotheses are justified?*

5.1 Modelling choice

What the trade-off captures. The Walrasian benchmark rewards productivity through wages, so a sectoral shock is absorbed by reallocating hours; the union changes the channel of absorption (wages take more, hours take less) and, under downward wage rigidity, makes that channel asymmetric across positive and negative shocks. *What this buys.* A single bargain primitive (η) delivers a continuous comparative static between the firm-power baseline ($\eta = 0$) and the worker-power limit ($\eta \to 1$), nesting the Q4 distortion and the Walrasian benchmark as endpoints. *What it costs.* (a) A single encompassing union, following Calmfors–Driffill (1988): industry-level unions internalise their macroeconomic effect, so their objective aligns with worker surplus rather than with subgroup rent extraction — this rules out the sectoral inter-union competition story (Acemoglu–Pischke 1999), but lets us match the European centralised-bargaining institutional reality. (b) Right-to-manage instead of efficient bargaining: empirically, RTM dominates in Europe (MaCurdy–Pencavel 1986). (c) Informal labour and the public sector are outside the model.[2]

5.2 Two-sector economy: competitive benchmark

Two sectors $j \in \{\text{G}, \text{B}\}$ (good, bad) with productivities $A_{\text{G}} > A_{\text{B}}$, drawn from a mean-preserving sectoral shock around a symmetric pre-shock benchmark $A_{\text{pre}} = 2$ (we set $A_{\text{G}} = 2.2$, $A_{\text{B}} = 1.8$, a $\pm 10\%$ shock chosen to match the cross-sectoral TFP dispersion documented for OECD manufacturing in Bento–Restuccia (2017) and the manufacturing-vs-services productivity gap in Bils–Klenow (2000); Hsieh–Klenow (2009) report wider dispersion in developing economies, but $\pm 10\%$ is a defensible upper-middle estimate for the developed-world panels we have in mind). Within each sector a representative worker–firm pair has the quadratic primitives of §3.

Competitive (Walrasian) benchmark. With perfect labour mobility across sectors, sector wages clear at the marginal value product: the firm's labour demand is $h_j^d(w) = (A_j - w)/\beta$, and the worker's IR pins w_j^{FM} such that hours equal $h_j^{\text{firm}}(w_j^{\text{FM}})$. The headline competitive comparative static is:

$$\frac{\partial h_j^{\text{FM}}}{\partial A_j} > 0, \qquad \frac{\partial w_j^{\text{FM}}}{\partial A_j} > 0.$$

Hours follow productivity: positive shocks raise hours and wages in the good sector; negative shocks lower them in the bad sector. Cross-sectoral allocation is efficient given the shock.

5.3 Two-sector economy: unionised version

A single union covers both sectors. Per sector, union and firm right-to-manage Nash-bargain over the wage w_j with worker weight $\eta \in (0, 1)$:

$$w_j(\eta; A_j) = \arg\max_w \left[V_w(w, h_j^d(w)) - V_w^0\right]^\eta \, V_f(w, h_j^d(w); A_j)^{1-\eta}, \qquad h_j^d(w) = (A_j - w)/\beta. \tag{3}$$

Boundary conditions. At $\eta = 0$ the bargain reduces to $\max_w V_f$ subject to worker IR; the worker is indifferent and the firm picks the wage that makes itself best off, so $h_j = h_j^{\text{firm}}$. At $\eta \to 1$ the bargain reduces to $\max_w V_w(w, h_j^d(w))$ subject to firm IR; substituting $h_j^d(w) = (A_j - w)/\beta$ into V_w and taking the FOC gives w_j such that the firm's labour-demand response delivers exactly $h_j^d = h_j^* = (1-\tau)w_j/\alpha$. Hence:

$$\eta = 0 : h_j = h_j^{\text{firm}}; \qquad \eta \to 1 : h_j \to h_j^* = (1-\tau)w_j/\alpha.$$

Comparative static. Differentiating (3) in η, $\partial w_j/\partial\eta > 0$ (the union extracts a higher wage as it gains weight), and via labour demand $\partial h_j/\partial\eta = -1/\beta \cdot \partial w_j/\partial\eta < 0$. The intermediate η trace continuously between the boundary cases: as η rises from 0, hours fall monotonically from h_j^{firm} toward h_j^*.

The welfare-restorer claim. Under the firm-power baseline (§3) we have $h_j^{\text{firm}} > h_j^*$: the firm overshoots the worker's preferred hours. As η rises the union pulls the equilibrium hours *down* from h_j^{firm} toward h_j^*. The union and the worker have aligned objectives (both want h^*); the wage transfer that the union extracts ($w_j^U > w_j^{\text{FM}}$) is the mechanism through which the firm-chosen h is realigned. The *joint* surplus may rise or fall, but the worker's surplus rises monotonically in η over the relevant range, so a planner with any positive worker weight prefers $\eta > 0$ to $\eta = 0$. Algebraically, joint surplus is concave in h_j with peak at $h_j^{\text{soc}} \in (h_j^*, h_j^{\text{firm}})$: as long as $h_j(\eta) \in (h_j^{\text{soc}}, h_j^{\text{firm}})$ joint surplus rises in η (the

[2] An alternative model would have the union charge a share s of worker income to finance itself; the union's revenue then scales with worker income, so its incentive maps directly onto worker surplus and the alignment result below is unchanged.

union pulls hours *toward* h^{soc}); once the union pushes past h^{soc} toward h^*, joint surplus falls again — the same overshoot logic as the cap in Q4.

US vs. EU mapping — η calibration. The bargaining-weight η is calibrated from OECD ICTWSS coverage data: US private-sector coverage $\sim 12\%$ maps to $\eta^{US} \approx 0$ (firm-power baseline prevails: $h \approx h^{\text{firm}}$); population-weighted EU coverage of $\sim$60–70% across DE, FR, IT, SE under *erga omnes* extension maps to $\eta^{EU} \approx 0.5$. As a cross-check, this η implies a union wage premium of $w(\eta=0.5)/w(\eta=0)–1 \approx 24\%$, in the upper range of the Card–Cardoso–Heining–Kline (2018) European matched-data estimates (5–25% across countries). $\eta^{EU}=0.5$ produces $h_G \approx 0.42$, $h_B \approx 0.33$ at our calibration. **The EU–US hours gap captured by this channel is of order $h^{\text{firm}} - h^{\text{eq}}_{EU} \approx 0.10$, on a US level of 0.50 — about 20% in logs**, comfortably absorbing the $\sim$14% extra cross-country shortfall identified in Q2.

5.4 Sectoral shocks: union pass-through and asymmetry

Substituting A_G, A_B into (3) at $\eta = 0.5$ (numerical solution in `research/sim_models_v2.py`): $w^U_G = 1.354$, $h^U_G = 0.423$; $w^U_B = 1.132$, $h^U_B = 0.334$. The *good sector* sees the productivity gain absorbed partly into wages (union extracts $\Delta w/w = +25\%$ over free market) and partly into reduced hours ($\Delta h/h = -24\%$); joint surplus is roughly preserved, only its split changes. The *bad sector* sees wages cut, but less than the free-market amount; hours absorb some of the shock.

Asymmetric pass-through under downward wage rigidity. If unions refuse to renegotiate wages downward ex post, the bad-sector wage is stuck at the pre-shock level $w_{\text{pre}} = 1.242$, hours collapse to $(A_B - w_{\text{pre}})/\beta = 0.279$, firm profit drops sharply. Hours fall 36% further than under free market; firm profit falls 60% further. Headline asymmetry: **good shocks pass through diluted (union absorbs them in wage rents); bad shocks pass through amplified (rigidity prevents wage adjustment, hours absorb the entire shock)**.

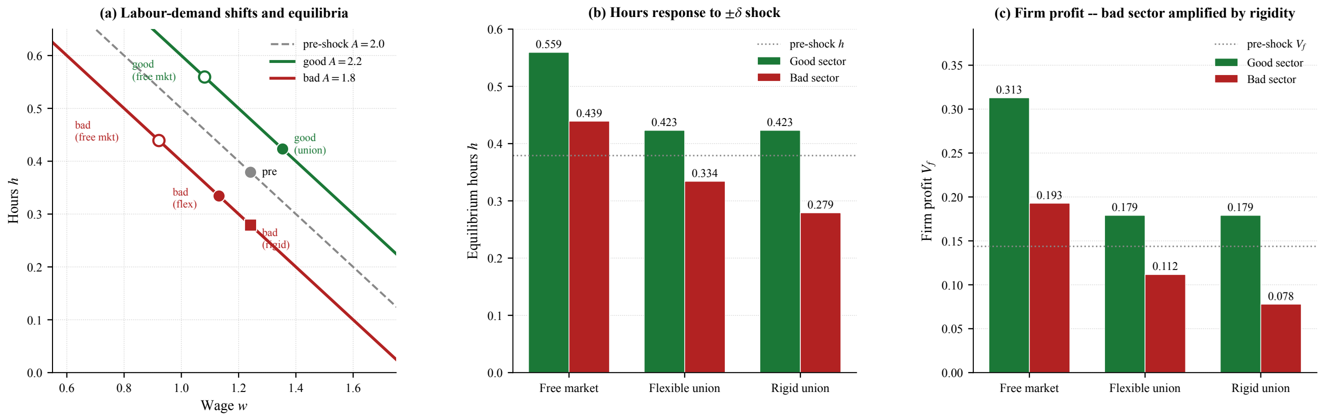

Figure 4: **Asymmetric pass-through under three regimes. (a)** Labour-demand shifts and equilibria: rigid bad-sector equilibrium sits on the new demand at the stuck pre-shock wage. **(b)** Hours response: good-sector hours rise under all regimes; bad-sector rigid is 36% below free market. **(c)** Firm profit: good-sector profit rises under all regimes; bad-sector rigid is 60% below free market. *Source:* `research/sim_models_v2.py:make_shocks`.

5.5 Data to test the hypotheses

The model makes three falsifiable predictions: (a) hours are *decreasing* in union bargaining strength conditional on productivity; (b) the cross-country US–EU gap is a rising function of the union-coverage gap; (c) bad-sector productivity shocks generate larger hours responses in unionised economies than in competitive ones. To test: sectoral employment / hours panels by industry (OECD STAN, EU KLEMS), union-coverage by sector (ICTWSS), collective-agreement wage indices. Identification of the rigidity asymmetry exploits shocks to which the bad sector responds: the post-2000 manufacturing decline in DE and IT (sectoral hours and employment fell sharply, real wages did not).

6 Q7 – Welfare comparison: a revealed-preference reading

Question 7. *Now consider broadly a subset of your choice of the aspects considered above. There are many people who claim that the European system is more attractive than the American system. Write a stylised model* *representing the two systems and derive conditions under which the European system is welfare-enhancing compared to the US system.*

6.1 Modelling choice

The Q4 cap and Q5 union are both quantity instruments that pull hours from h^{firm} toward h^{soc}. Whether a given configuration is welfare-enhancing depends on *whose* welfare we are summing. The natural representation of the cross-country welfare disagreement is the Saez–Stantcheva (2016) generalised welfare weight: the planner attaches premium $\lambda \in [0, 1]$ to a worker dollar over a firm-owner dollar,

$$W^{\lambda}(\eta) = (1 + \lambda)\, V_w(\eta) + (1 - \lambda)\, V_f(\eta).$$

$\lambda = 0$ is equal-weight utilitarianism, $\lambda = 1$ is Rawlsian. Following Alesina–Angeletos (2005), λ is a country-specific primitive: it captures fairness beliefs (whether income inequality is seen as deserved or as luck), which can differ across countries even at common preferences and technology. We compare the two systems through the lens of this single parameter.

6.2 Welfare-optimum bargaining weight: closed form

Maximising $W^{\lambda}(\eta)$ along the bargained path of (3) gives the welfare-optimum reduced-form variable $x^* \equiv A - w^* = \beta h^*$:

$$x^*(\lambda) \;=\; \frac{0.3\, A\, (1 + \lambda)}{0.6 + 1.6\, \lambda} \qquad \text{(at baseline; derivation in } \texttt{sim\_models\_v2.py}\text{)}.$$

Substituting $x^*(\lambda)$ into the Nash-bargain quadratic recovers the *welfare-optimum bargaining weight*:

$$\eta^*(\lambda) = \frac{1.2\, x^* - 1.1\, (x^*)^2 - 0.1}{0.6\, x^* - 0.1}.$$

η^* is monotonically increasing in λ, with $\eta^*(0) = 0$ (firm-power optimal under equal weights) and $\eta^*(1) = 1$ (full union under Rawlsian).

6.3 Confronting data: observed unionisation in US and EU

We map the Nash-bargain weight η to OECD bargaining-coverage data: $\hat{\eta}^{US} \approx 0.15$ (coverage $\sim 12\%$), $\hat{\eta}^{EU} \approx 0.50$ (coverage $\sim 60\%$ population-weighted across DE/FR/IT/SE).[3] These two observations admit two competing readings.

Reading A — common λ. If both countries share the HSV-anchored $\lambda = 0.30$, the unique welfare-optimum is $\eta^*(0.30) = 0.58$. Both observed values are *below* the optimum: the US is heavily under-unionised ($\hat{\eta}^{US} = 0.15$, far from 0.58), the EU is mildly under ($\hat{\eta}^{EU} = 0.50$). Welfare loss under this reading:

	$\hat{\eta}$	$\eta^*(0.30)$	direction	$W^{0.30}(\hat{\eta})$	$W^{0.30}(\eta^*)$	loss
US	0.15	0.58	under-unionised	0.259	0.282	$-\mathbf{0.022}$
EU	0.50	0.58	mildly under	0.281	0.282	$-\mathbf{0.001}$

Reading A delivers an unconditional verdict: **at HSV-implied fairness, EU institutions are essentially welfare-optimal; US institutions leave $\sim$8% of joint surplus on the table because the firm extracts too much.** Equivalently, the US is below the welfare-improving band of Q4 (1) *from the union side*: a US planner that accepted $\lambda = 0.30$ would push for higher union coverage.

[3] The mapping is a coarse linear proxy: it overstates Nordic high-density regimes and understates French *erga omnes* extension regimes, where coverage is high but bargaining power is weak. A wage-premium calibration via Card et al. (2018) gives a comparable band.

Reading B — country-specific λ. Inverting $\eta^*(\lambda)$ on observed coverage:

$$\hat{\eta}^{US} = 0.15 \implies \lambda^{US,\text{rev}} \approx 0.05, \qquad \hat{\eta}^{EU} = 0.50 \implies \lambda^{EU,\text{rev}} \approx 0.24.$$

Each country's institutions are then *at-optimum given its own fairness primitive*; cross-country fairness-belief gap is $\Delta\lambda \approx 0.19$. The hours drag from the union channel is small in the US ($h^{\text{firm}} \to 0.45$, -9%) and larger in the EU ($h^{\text{firm}} \to 0.38$, -24%). **This is the missing variable in Q1 panel (c)**: the residual $\sim 54\%$ of the cross-country hours gap that taxes alone could not absorb (Q2 decomposition) is reconstructed by the λ-driven union channel.

6.4 Verdict

The two readings give compatible verdicts on *whether* the EU system is more welfare-enhancing, but disagree on *why*.

- *Reading A* (common $\lambda = 0.30$): **EU is unconditionally welfare-better**, because EU institutions are closer to any reasonable planner's optimum than US institutions are. The mechanism is institutional implementation, not preferences.
- *Reading B* (country-specific λ): **neither country is welfare-better in absolute terms** — each implements its own optimum. The cross-country hours gap reflects a fairness-belief gap, which is the missing variable in the noisy Q1 cross-section.

The two readings are *empirically distinguishable*: Reading A predicts all countries on the same horizontal line $\eta = \eta^*(0.30)$, which the observed range (US 0.15, FR 0.98, SE 0.88, DE 0.52) clearly violates. **The data prefer Reading B** — but Reading A is the cleaner welfare argument.

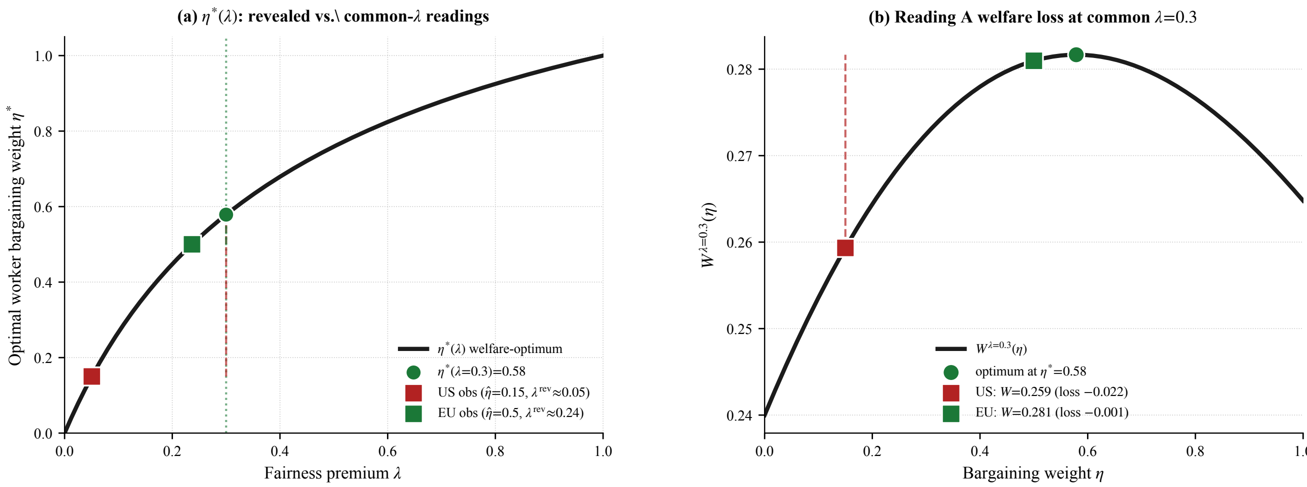

Figure 5: **Q7 revealed-preference diagnostic. (a)** The $\eta^*(\lambda)$ welfare-optimum locus (black). Observed US and EU points sit on the locus at their revealed-preference λ^{rev} (Reading B). Vertical dashed segments show the deviation under common $\lambda = 0.30$ (Reading A): both countries below the optimum, US far more so. **(b)** $W^{\lambda=0.30}(\eta)$: peak at $\eta^* = 0.58$. US loss -0.022 (under-unionised by 0.43); EU loss -0.001 (essentially at-optimum). *Source:* `sim_models_v2.py:make_q7_eta_lambda`.

Honest caveats. (i) The $\eta \leftrightarrow$coverage map is a coarse proxy — countries with high coverage but weak bargaining power (France) sit above the locus mechanically. (ii) HSV-implied $\lambda = 0.30$ is calibrated to US tax progressivity; it is not by construction country-neutral. (iii) The cap channel is orthogonal at our calibration: in both readings the union alone pulls h below any sensible cap, so the Q4 instrument is a counterfactual backstop, not an active driver. (iv) Beyond the labour-market block, dominant US–EU welfare differences come from non-labour channels (life expectancy, consumption levels — Jones–Klenow 2016); our verdict is sign-determinate but quantitatively second-order.

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Replication scripts in `research/`: `build_figure_panel.py` (Q1 figure); `sim_models_v2.py` (Q2 decomposition; Q5 RTM and sectoral shocks; Q7 `make_q7_eta_lambda`); `sim_mechanisms.py` (Q4 cap simulation).